



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to M. BADRI FUEL STATION

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

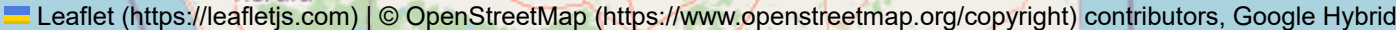
## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





End: (12.407136, 78.145978)



# Welcome to the Journey Risk Management Study

## Route Overview:

The route covers approximately 148.23 kilometers, starting from Sangagiri and ending in Morana halli, traversing through key checkpoints along major roads such as Sankari RS - Erode Privu Rd, Bhavani Main Road, and SH 86.

### 1. Typical Weather Conditions:



- Tamil Nadu generally experiences tropical weather, with the dry season from March to June, the monsoon from July to September, and the post-monsoon from October to December. Heavy rains can cause flooding, affecting visibility and road conditions.

## 2. Traffic Patterns:

- Major congestions may occur during morning and evening peak hours, especially near urban areas like Sangagiri and Erode. Expect congestion on SH 86 as this is a critical connector road.

## 3. Road Quality and Infrastructure:

- While the National and State Highways are generally well-maintained, watch for potential potholes and uneven road surfaces in more rural sections. Roads can narrow considerably through towns, demanding cautious driving.

## 4. Alternative Route Suggestions:

- An alternative to consider in case of emergency is transitioning from SH 86 to NH 44 at Dharmapuri, depending on traffic predictions and current conditions.

## 5. Local Regulations on Hazardous Material Transport:

- Familiarity with Tamil Nadu's specific regulations for the transport of hazardous materials is crucial. Adhering to designated routes and following all signage is mandatory. Overnight stops in populated areas may be restricted for safety reasons.

## 6. History of Incidents:

- Historically, some minor incidents have involved heavy vehicles due to overloading, especially on Bhavani Main Road. It is important to adhere strictly to load limits.

## 7. Environmental and Sensitive Areas:

- There are several ecologically sensitive zones along the route, especially near forested areas. Minimizing noise and pollution is essential, and additional caution is required in these parts.

## 8. Communication Coverage:

- While urban areas generally have good network coverage, rural stretches, especially between Erumaipatti and Kaveripattinam, may experience dead zones.

## 9. Emergency Response Times:

- Emergency response times can vary. Urban areas are typically within 30-40 minutes' reach of emergency services, but more rural areas might experience longer response times, up to an hour or more.

## 10. Overall Risk Assessment Summary:

- This route is moderately risky, primarily due to weather-related challenges during the monsoon, potential traffic congestion, and occasional infrastructure issues. Preparation for variable conditions, emergency contingencies, and strict adherence to regulations will minimize risks significantly.

In conclusion, while the route is largely manageable, careful planning, adherence to local transport regulations, and continuous monitoring of weather and traffic conditions can significantly mitigate potential risks for the transportation of hazardous materials.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.45352, 77.85698 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.45764, 77.85219 | 30 KM/Hr    |
| 5  | Turn       | High       | 11.46303, 77.84945 | 15 KM/Hr    |
| 6  | Turn       | High       | 11.45571, 77.81702 | 15 KM/Hr    |
| 7  | Turn       | High       | 11.45615, 77.81677 | 15 KM/Hr    |
| 8  | Turn       | Medium     | 11.46916, 77.82431 | 30 KM/Hr    |
| 9  | Blind Spot | Blind Spot | 11.49395, 77.88537 | 10 KM/Hr    |
| 10 | Turn       | High       | 11.49465, 77.88488 | 15 KM/Hr    |
| 11 | Turn       | Medium     | 11.56096, 77.89931 | 30 KM/Hr    |
| 12 | Turn       | Medium     | 11.59214, 77.90656 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.63833, 77.94818 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.64171, 77.95184 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.64243, 77.95198 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.64721, 77.95118 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.67870, 77.96846 | 30 KM/Hr    |
| 18 | Turn       | High       | 11.70227, 77.97864 | 15 KM/Hr    |
| 19 | Turn       | Medium     | 11.71715, 77.99969 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.71921, 78.01133 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.72737, 78.02248 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.73775, 78.03465 | 30 KM/Hr    |
| 23 | Turn       | High       | 11.73651, 78.03878 | 15 KM/Hr    |
| 24 | Turn       | Medium     | 11.74189, 78.04660 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.74309, 78.04725 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.78039, 78.05792 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.86753, 78.08331 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.86981, 78.08387 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 29 | Turn       | Medium     | 11.88041, 78.08143 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.91302, 78.05856 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.95625, 78.04748 | 30 KM/Hr    |
| 32 | Turn       | High       | 11.95880, 78.04989 | 15 KM/Hr    |
| 33 | Turn       | High       | 11.96040, 78.04947 | 15 KM/Hr    |
| 34 | Turn       | High       | 11.96320, 78.05139 | 15 KM/Hr    |
| 35 | Turn       | Medium     | 11.96463, 78.05077 | 30 KM/Hr    |
| 36 | Turn       | Medium     | 11.96504, 78.04975 | 30 KM/Hr    |
| 37 | Turn       | High       | 11.96597, 78.04920 | 15 KM/Hr    |
| 38 | Turn       | Medium     | 11.96679, 78.04942 | 30 KM/Hr    |
| 39 | Turn       | Medium     | 11.96804, 78.05080 | 30 KM/Hr    |
| 40 | Turn       | Medium     | 11.98504, 78.06287 | 30 KM/Hr    |
| 41 | Turn       | Medium     | 12.24590, 78.19512 | 30 KM/Hr    |
| 42 | Turn       | Medium     | 12.25185, 78.20031 | 30 KM/Hr    |
| 43 | Turn       | Medium     | 12.25477, 78.20000 | 30 KM/Hr    |
| 44 | Blind Spot | Blind Spot | 12.42228, 78.21903 | 10 KM/Hr    |
| 45 | Turn       | Medium     | 12.42287, 78.21300 | 30 KM/Hr    |
| 46 | Turn       | Medium     | 12.40299, 78.16057 | 30 KM/Hr    |
| 47 | Turn       | Medium     | 12.40266, 78.15981 | 30 KM/Hr    |
| 48 | Blind Spot | Blind Spot | 12.40556, 78.14583 | 10 KM/Hr    |

Emergency Locations

|   | type     | name                       | coordinates            | speed_limit | risk_level |
|---|----------|----------------------------|------------------------|-------------|------------|
| 1 | hospital | Sri Shanthi Hospital       | 11.569272, 77.901937   | 30 km/h     | Medium     |
| 2 | hospital | Sugumar Hospital           | 11.7396031, 78.0426955 | 30 km/h     | Medium     |
| 3 | hospital | Omalur Government Hospital | 11.74161, 78.044296    | 30 km/h     | Medium     |
| 4 | clinic   | R.R. Nursing Home          | 11.7394129, 78.0439861 | 30 km/h     | Medium     |
| 5 | hospital | Sri Hari Hospital          | 11.7396726, 78.0440701 | 30 km/h     | Medium     |
| 6 | hospital | Sri Sugam Hospital         | 11.7429821, 78.0461039 | 30 km/h     | Medium     |
| 7 | hospital | Saravana Hospital          | 11.7409303, 78.0458308 | 30 km/h     | Medium     |
| 8 | hospital | Arun Hospital              | 11.7429892, 78.0466881 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates            | speed_limit | risk_level |
|----|----------|------------------------------------|------------------------|-------------|------------|
| 9  | clinic   | Saratha Clinic                     | 11.7422818, 78.0458175 | 30 km/h     | Medium     |
| 10 | clinic   | Gomathi Clinic                     | 11.7445293, 78.0470401 | 30 km/h     | Medium     |
| 12 | clinic   | Sri Meenakshi Nursing Home, Salem  | 11.7449162, 78.0474637 | 30 km/h     | Medium     |
| 13 | clinic   | Divyam Child Care Clinic           | 11.7447135, 78.0473858 | 30 km/h     | Medium     |
| 14 | clinic   | Momota Clinic                      | 11.744982, 78.047397   | 30 km/h     | Medium     |
| 15 | hospital | Amirthalingam Hospital             | 11.7456708, 78.0470548 | 30 km/h     | Medium     |
| 16 | hospital | Vinayaka Child Hospital            | 11.745812, 78.0480225  | 30 km/h     | Medium     |
| 17 | hospital | Dr. G.R.Thilakam Hospital          | 11.746097, 78.04769    | 30 km/h     | Medium     |
| 18 | hospital | Dr. Nagaraj Hospital               | 11.7464035, 78.0472401 | 30 km/h     | Medium     |
| 19 | hospital | Thulasi Hospital                   | 11.7461516, 78.0469984 | 30 km/h     | Medium     |
| 20 | hospital | Children Hospital                  | 11.7458377, 78.0480205 | 30 km/h     | Medium     |
| 21 | clinic   | Shri Ramana Poly Clinic            | 11.7477122, 78.0479293 | 30 km/h     | Medium     |
| 22 | clinic   | Manimuthu Skin Clinic              | 11.7521189, 78.0484058 | 30 km/h     | Medium     |
| 24 | hospital | Anugraha Multi Speciality Hospital | 11.773549, 78.0546408  | 30 km/h     | Medium     |
| 26 | clinic   | JSA Clinic                         | 11.8700619, 78.0844982 | 30 km/h     | Medium     |
| 27 | hospital | L.R. Hospital                      | 11.8703354, 78.0844554 | 30 km/h     | Medium     |
| 28 | hospital | Thoppur,Government Hospital        | 11.9398824, 78.0551658 | 30 km/h     | Medium     |
| 30 | hospital | Government Hospital                | 12.3048002, 78.2060881 | 30 km/h     | Medium     |

Crowded Spots

|    | type    | name                                              | coordinates            | speed_limit | risk_level |
|----|---------|---------------------------------------------------|------------------------|-------------|------------|
| 0  | school  | KRP Matric. Hr. Sec School                        | 11.4546193, 77.8142445 | 30 km/h     | Medium     |
| 11 | school  | Fathima Hr Sec School                             | 11.7410508, 78.0479272 | 30 km/h     | Medium     |
| 23 | school  | John Britto Matriculation Higher Secondary School | 11.7676907, 78.0513092 | 30 km/h     | Medium     |
| 25 | college | Dhirajlal Gandhi College of Technology            | 11.7968014, 78.0562095 | 30 km/h     | Medium     |
| 29 | college | Laxminarayana Polytechnic College                 | 12.0421277, 78.103203  | 30 km/h     | Medium     |



|    | type   | name                                    | coordinates               | speed_limit | risk_level |
|----|--------|-----------------------------------------|---------------------------|-------------|------------|
| 31 | school | Government Boys Higher secondary school | 12.3047059,<br>78.2065495 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45764, 77.85219





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45571, 77.81702



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45615, 77.81677





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46916, 77.82431



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.49395, 77.88537





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49465, 77.88488



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56096, 77.89931



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.59214, 77.90656





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.63833, 77.94818



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64171, 77.95184





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64243, 77.95198



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64721, 77.95118



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.67870, 77.96846





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.70227, 77.97864



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71715, 77.99969





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71921, 78.01133



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.72737, 78.02248



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.73775, 78.03465





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.73651, 78.03878



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.74189, 78.04660





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.74309, 78.04725



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.78039, 78.05792



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.86753, 78.08331





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.86981, 78.08387



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.88041, 78.08143





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.91302, 78.05856



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95625, 78.04748



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.95880, 78.04989

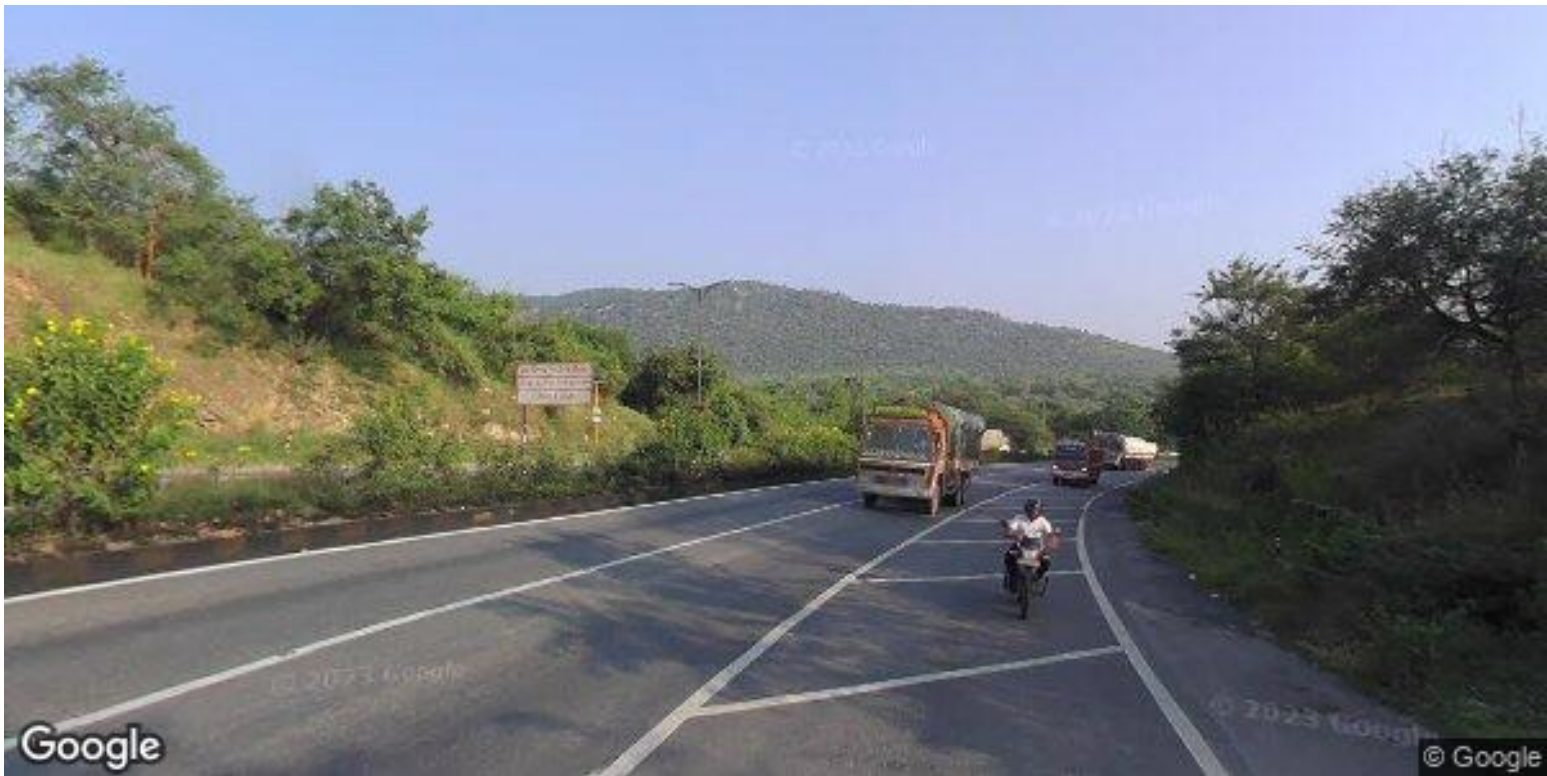




**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96040, 78.04947



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96320, 78.05139





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96463, 78.05077



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96504, 78.04975



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96597, 78.04920





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96679, 78.04942



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96804, 78.05080





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.98504, 78.06287



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.24590, 78.19512



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.25185, 78.20031





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.25477, 78.20000



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 12.42228, 78.21903





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.42287, 78.21300

